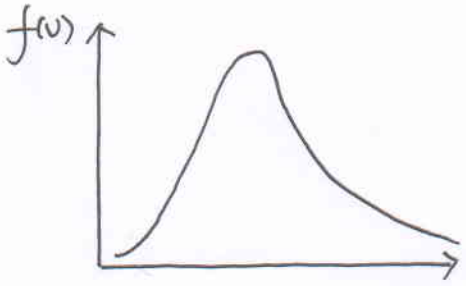
麦克斯韦分布的积分技巧



$$f(v) = Cv^2 e^{-\frac{\frac{1}{2}mv^2}{k_B T}} \quad \text{Let } a \equiv \frac{m}{2k_B T}$$

$$= Cv^2 e^{-av^2}.$$

注意高斯积分有: $\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$ $\therefore \int_0^{\infty} e^{-ax^2}dx = \frac{\sqrt{\pi}}{2\sqrt{a}}$

而 $I(2) = \int_0^{\infty} x^2 e^{-ax^2}dx = -\int_0^{\infty} \frac{\partial}{\partial a}(e^{-ax^2})dx = -\frac{\partial}{\partial a}\int_0^{\infty} e^{-ax^2}dx$

$$= -\frac{\partial}{\partial a}\left[\frac{\sqrt{\pi}}{2\sqrt{a}}\right] = \frac{\sqrt{\pi}}{4a^{\frac{3}{2}}}.$$

附: 如果要计算 C (举例来说), 可根据归一化条件.

$$\int_0^{\infty} f(v)\,dv = 1$$

左边 $I = \int_0^{\infty} f(v)\,dv = \int Cv^2 e^{-av^2}dv = C\frac{\sqrt{\pi}}{4a^{\frac{3}{2}}} = \text{右边} = 1$

$$\therefore C = \frac{4a^{\frac{3}{2}}}{\sqrt{\pi}} = 4\pi\left(\frac{m}{2k_B T\pi}\right)^{\frac{3}{2}} \#$$

进一步地, (同学们自己去推导得出)

$$I(0) = \int_0^{\infty} x^0 e^{-ax^2}dx = \frac{\sqrt{\pi}}{2\sqrt{a}}$$

$$I(1) = \int_0^{\infty} x^1 e^{-ax^2}dx = \frac{1}{2a}$$

$$\vdots$$

$$I(n) = -\frac{\partial}{\partial a} I(n-2)$$